DSP - LEC – 4 - Ch-1

CHAPTER – 1

1.6 Analog-to-Digital Conversion

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1.6 Analog-to-Digital Conversion

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1.6 ANALOG-TO-DIGITAL CONVERSION

- Most signals of practical interest, such as speech, biological signals, seismic signals, radar signals, sonar signals, and various communications signals such as audio and video signals, **are analog.**

- To process analog signals by digital means, it is first necessary to convert them into digital form**, i.e.** to convert them to a sequence of number having finite precision. This procedure is called A / D conversion.

Sec. 1.6 Cont. - 2

- We view A / D conversion as a three-step process (**Fig. 1.9**).

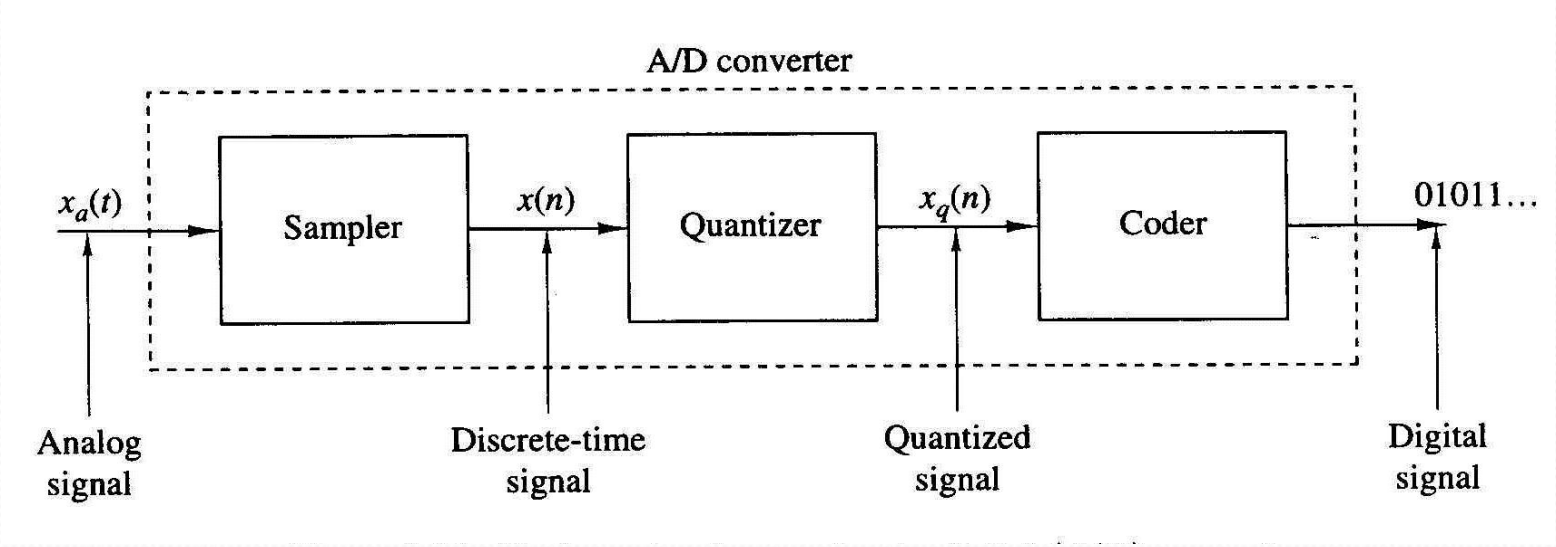


Fig. 1.9 Basic parts of an analog-to-digital (A/D) converter.

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(1) Sampling

- This is the conversion of a continuous-time signal into a discrete-time signal obtained by taking “samples” of the continuous-time signal at discrete-time instants.

- Thus, if $x_a(t)$ is the input to the sampler, the output is $x_a(nT)$ $x(n)$, where T is called the sampling interval.

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(2) Quantization

- This is the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each signal sample is represented by a value selected from a finite set of possible values.

- The difference between the un-quantized sample x(n) and the quantized output $x_q(n)$ is called the quantization error.

(3) Coding

- In this process, each discrete value $x_q(n)$ is represented by a b-bit binary sequence.

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In particular, we demonstrate that sampling does not result in a loss of information, nor does it introduce distortion in the signal if the signal bandwidth is finite.

So, we say that sampling process is invertible or reversible process.

On the other hand, quantization is a noninvertible or irreversible process that results in signal distortion.

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1.6.1 Sampling of Analog Signals

- There are many ways to sample an analog signal. We limit our discussion to *periodic or uniform sampling*, which is the type of sampling used most often in practice.
- This is described by the relation:

$$x(n) = x_a(nT), \; -\infty < n < \infty \qquad (1.29)$$

where x(n) is the discrete-time signal obtained by "taking samples" of the analog signal xa(t) every T seconds. This procedure is illustrated in **Fig. 1.11**.

The time interval T between successive samples is called the *sampling period* or *sample interval* and its reciprocal 1/T = Fs is called the *sampling rate* (samples per second) or the *sampling frequency* (hertz).

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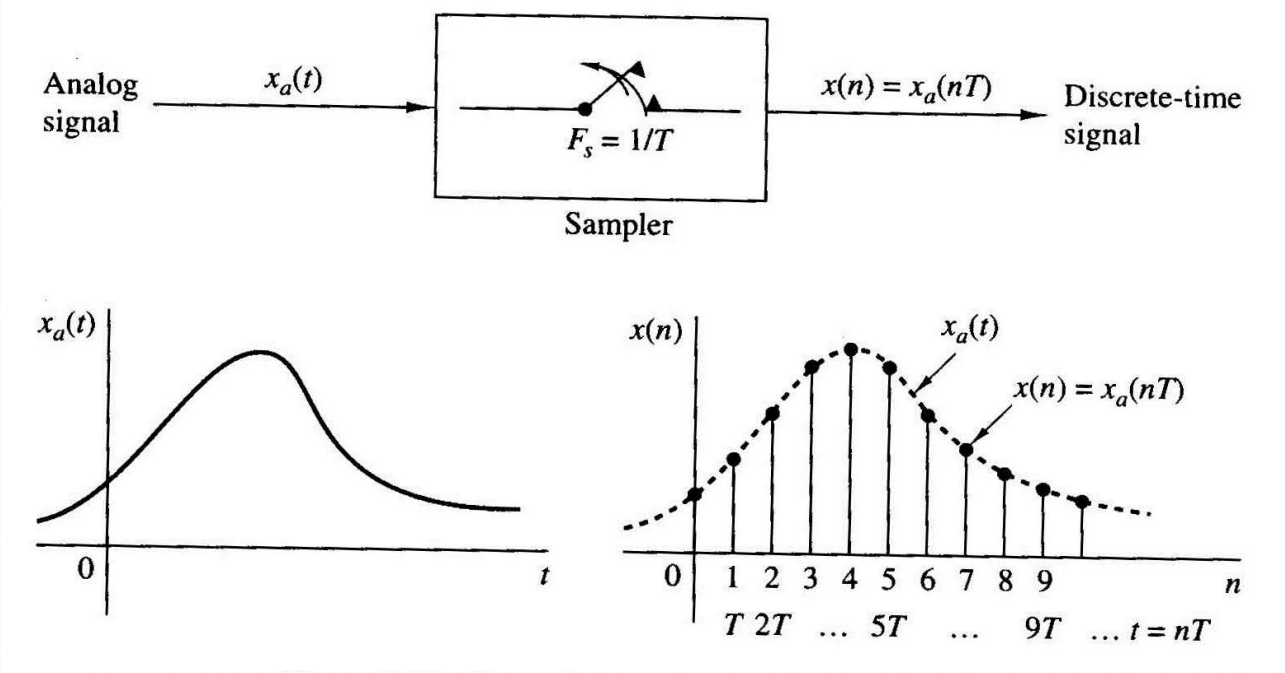


Fig. 1.11 Periodic sampling of an analog signal.

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- Periodic sampling establishes a relationship between the time variables‘t’ and ‘n’ of continuous-time and discrete-time signals, respectively.

- Indeed, these variables are linearly related through the sampling period T or, equivalently, through the sampling rate $F_s = 1/T$, as:

$$t = nT = \frac{n}{F_s} \qquad (1.30)$$

- As a consequence of (1-30), there exists a relationship between the frequency variable f (or ω) for analog signals and the frequency variable F (or Ω) for discrete-time signals.

Sec. 1.6 Cont. - 9

- To establish this relationship, consider an analog sinusoidal signal of the form:

$$x_a(t) = A \cos (2\pi f t + \theta) \qquad (1.31)$$

which, when sampled periodically at a rate $F_s = 1/T$ samples per second, yields

$$x_a(nT) = x(n) = A \cos (2\pi f n T + \theta)$$

$$= A \cos \left(\frac{2\pi n f}{F_s} + \theta\right) \qquad (1.32)$$

$$x(n) = A \cos (2\pi F n + \theta),\ -\infty < n < \infty \qquad (1.14)$$

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If we compare **(1.32)** with **(1.14)**, we note that the frequency variables F and f are linearly related as

$$F = \frac{f}{F_s} \tag{1.33}$$

or, equivalently, as

$$\Omega = \omega T \tag{1.34}$$

As (1.33) implies, we can use “F” to determine the frequency “f” in hertz only if the sampling frequency, F_s is known.

Sec. 1.6 Cont. - 11

We recall from section 1.5.1 (continuous-time signal) that the range of the frequency variable "f" or "ω" for continuous-time sinusoids are:

$$-\infty < f < \infty$$
$$-\infty < \omega < \infty \qquad (1.35)$$

However, the situation is different for discrete-time sinusoids. From section 1.5.2 (discrete-time signal) we recall that:

$$-\frac{1}{2} < F < \frac{1}{2}$$
$$-\pi < \Omega < \pi \qquad (1.36)$$

Sec. 1.6 Cont. - 12

By substituting from **(1.33) and (1.34)** into **(1.36),** we find that "f" when sampled at a rate $F_s = 1/T$ must fall the range:

$$-\frac{1}{2T} = -\frac{F_s}{2} \le f \le \frac{F_s}{2} = \frac{1}{2T} \qquad (1.37)$$

or, equivalently,

$$-\frac{\pi}{T} = -\pi F_s \le \omega \le \pi F_s = \frac{\pi}{T}$$

Sec. 1.6 Cont. - 13

Since the highest frequency in a discrete-time signal is $\Omega = \pi$ **or** $F = \frac{1}{2}$, it follows that, with a sampling rate F_s, the corresponding highest values of **"f"** and **"ω"** are:

$$f_{max} = \frac{F_s}{2} = \frac{1}{2T} \qquad (1.38)$$

$$\omega_{max} = \pi F_s = \frac{\pi}{T} \qquad (1.39)$$

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These relations are summarized in **Table 1.1.**

Table 1.1 Relations among Frequency Variables

Continuous-time signals	Discrete-time signals
$\omega = 2\pi f$	$\Omega = 2\pi F$
$\frac{radians}{\sec}$ Hz	$\frac{radians}{sample}$ $\frac{cycles}{sample}$
$\Omega = \omega T, F = f / F_s$	$-\pi \le \Omega \le \pi$ $-\frac{1}{2} \le F \le \frac{1}{2}$
$\omega = \Omega / T, f = F . F_s$	
$-\infty < \omega < \infty$	$-\pi / T \le \omega \le \pi / T$
$-\infty < f < \infty$	$-F_s / 2 \le f \le F_s / 2$

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1.6.2 sampling Theorem

- Given any analog signal, how should we select the sampling period "T" or, equivalently, the sampling rate F_s?

- To answer this question, we must have some information about the characteristics of the signal to be sampled.

- In particular, we must have some general information concerning the frequency content of the signal.

- Such information is generally available to us.
- **For example**, we know generally that the major frequency components of a speech signal fall below 3000 Hz.
  On the television signals, in general, contains important frequency components up to 5 MHz.

Sec. 1.6 Cont. - 16

- The amplitudes, frequencies, and phases usually change slowly with time from one time segment to another.

- However, suppose that the frequencies do not exceed some known frequency, say f_{max}.

- **For example:**

f_{max} = 3000 Hz for the class of speech signals and
f_{max} = 5 MHz for television signals.

Since the maximum frequency may vary slightly from different realizations among signals of any given class ***(e.g., it may vary slightly from speaker to speaker)***, we may wish to ensure that f_{max} does not exceed some predetermined value by passing the analog signal through a filter **that severely attenuates frequency components above f_{max}.**

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- Thus to avoid the problem of aliasing, F_s is selected so that

$$F_s > 2f_{max} \tag{1.47}$$

where f_{max} is the largest frequency component in the analog signal.

With the sampling rate selected in this manner, any frequency component, say $|f_i| < f_{max}$, in the analog signal is mapped into a discrete-time sinusoid with a frequency:

$$-\frac{1}{2} \le F_i = \frac{f_i}{F_s} \le \frac{1}{2} \tag{1.48}$$

or, equivalently,

$$-\pi \le \Omega_i = 2\pi F_i \le \pi \tag{1.49}$$

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Sampling theorem:

If the highest frequency contained in an analog signal $x_a(t)$ is $f_{max} = B$ and the signal is sampled at a rate $F_s > 2f_{max} = 2B$, then $x_a(t)$ can be exactly recovered from its sample values using the interpolation function (1.50):

$$g(t) = \frac{\sin 2\pi Bt}{2\pi Bt} \qquad (1.50)$$

- Thus $x_a(t)$ may be expressed as

$$X_a(t) = \sum_{n=-\infty}^{\infty} x_a\left(\frac{n}{F_s}\right) g\left(t-\frac{n}{F_s}\right) \qquad (1.51)$$

where $x_a(n/F_s) = x_a(nT) = x(n)$ are the samples of $x_a(t)$.

Sec. 1.6 Cont. - 19

- When the sampling of $x_a(t)$ is performed at the minimum sampling rate $F_s = 2B$, the reconstruction formula in **(1-51)** becomes

$$x_a(t) = \sum_{n=-\infty}^{\infty} X_a\left(\frac{n}{2B}\right) \frac{\sin 2\pi B(t - n/2B)}{2\pi B(t - n2B)} \qquad (1.52)$$

- The sampling rate $f_N = 2B = 2f_{max}$ is called the **Nyquist rate**.

- Because of the complexity and the infinite number of samples required in
- **(1.51) or (1.52)**, these reconstruction formulas are primarily of theoretical interest.

Sec. 1.6 Cont. - 20

- **Figure 1.14** illustrates the ideal D/A conversion process using the interpolation function in **(1-50).**

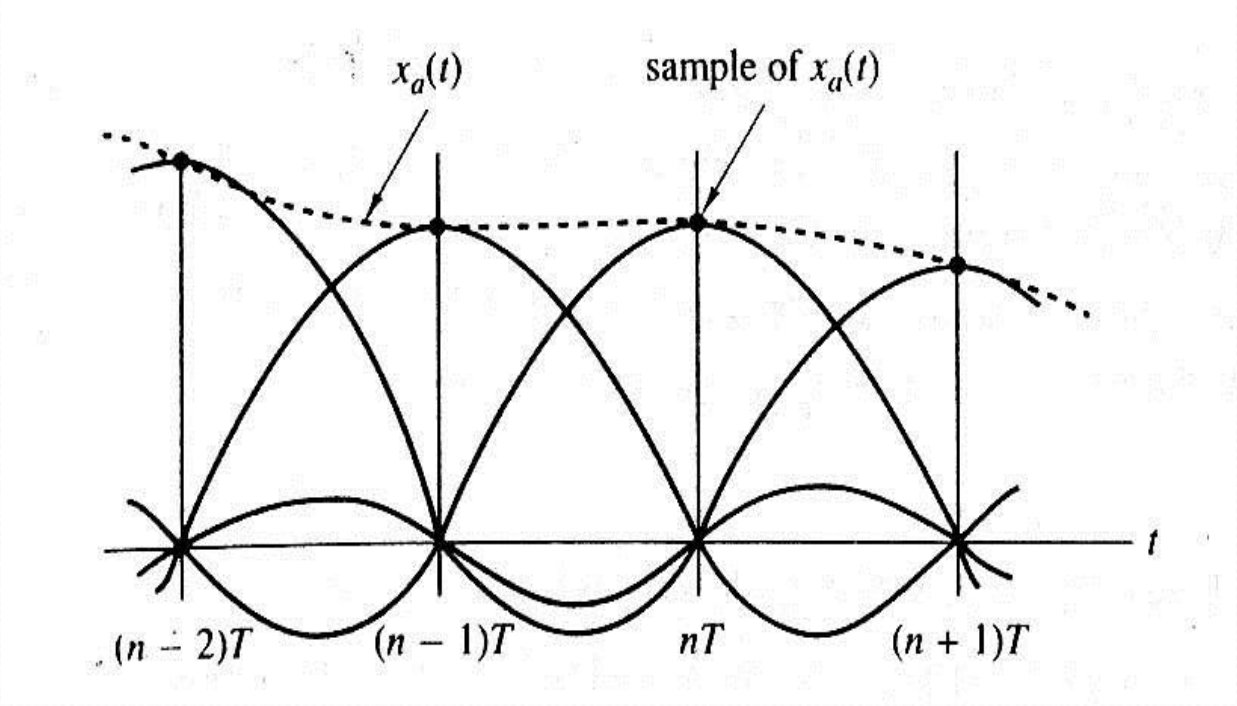


Fig. 1.14 Ideal D/A conversion (interpolation).

Sec. 1.6 Cont. - 21

1.6.3 Quantization of continuous-amplitude signals

- As we have seen, a digital signal is a sequence of numbers (samples) in which each number is represented by a finite number of digits (finite precision).

- The process of converting a discrete-time continuous-amplitude signal into a digital signal by expressing each sample value as a finite (instead of an infinite) number of digits, is called *quantization.*

- The error introduced in representing the continuous-valued signal by a finite set of discrete value levels is called *quantization error* or *quantization noise.*

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- We denote the quantizer operation on the samples **x(n)** as **Q[x(n)]** and

- **$x_q(n)$** … denote the sequence of quantized samples at the output of the quantizer.

Hence

$$x_q(n) = Q\,[x(n)]$$

- Then the quantization error is a sequence **$e_q(n)$** defined as the difference between the quantized value and the actual sample value.

Thus

$$e_q(n) = x_q(n) - x(n) \qquad (1.53)$$

Sec. 1.6 Cont. - 23

- We illustrate the quantization process by the **following example:**

- Let us consider the discrete-time signal: $x(n) = \begin{cases} 0.9^n, n \geq 0 \\ 0, n < 0 \end{cases}$

obtained by sampling the analog exponential signal: $x_a(t) = 0.9^t$, $t \geq 0$

with a sampling frequency Fs = 1 Hz (Ts= 1Sec), see **Fig. 1.15(a).**

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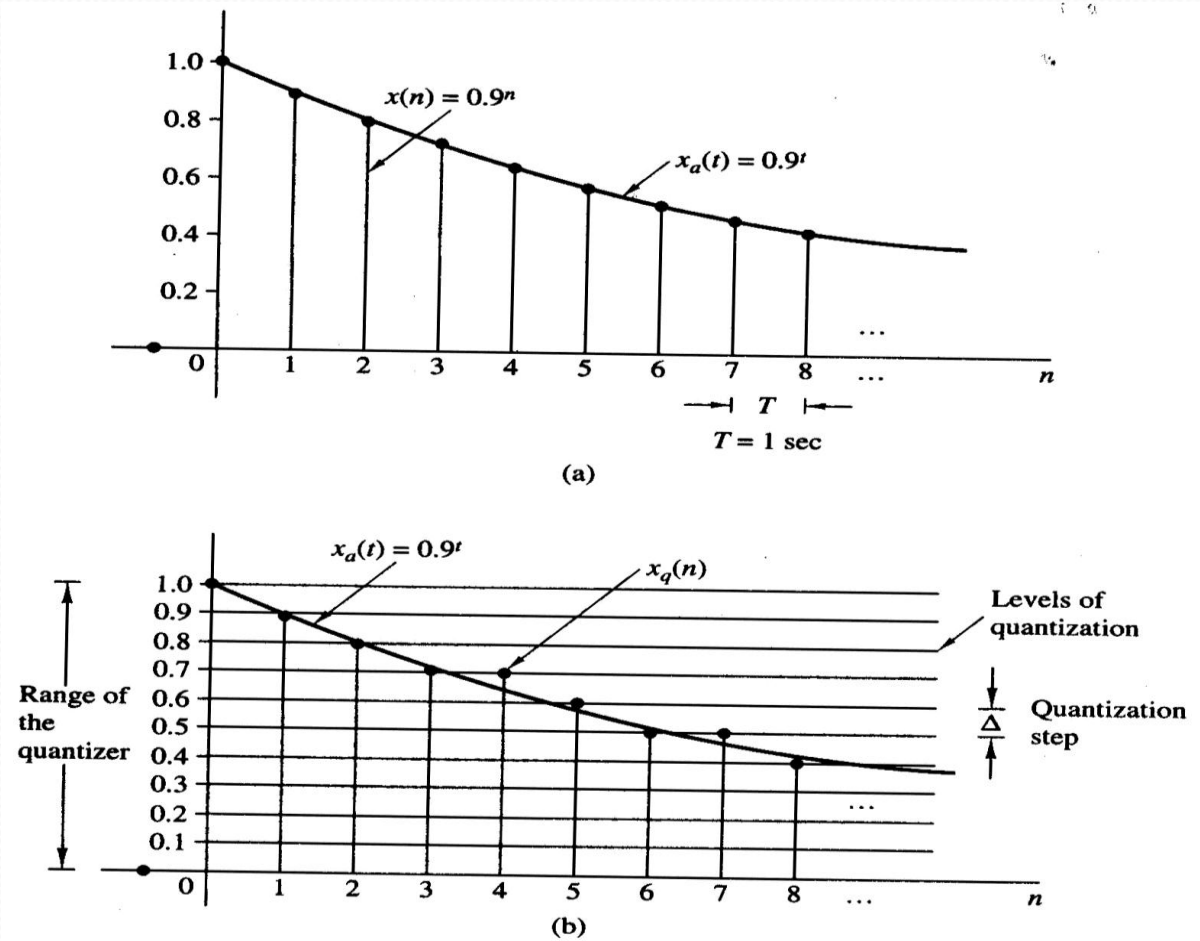


Fig. 1.15 Illustration of quantization

Sec. 1.6 Cont. - 25

- Observation of **table 1.2**, which shows the values of the first 10 samples of x(n), reveals that the description of the sample value x(n) requires n significant digits.

- However, let us assume that we want to use only one significant digit.
- To eliminate the excess digits, we can either simply discard them *(truncation)* or discard them by rounding the resulting number *(rounding)*.
- The resulting quantized signals $x_q(n)$ are shown in **Table 1.2.**

- We discuss only quantization by rounding, although it is just as easy to treat truncation. The rounding process is graphically illustrated in **Fig. 1.15b**.

- The values allowed in the digital signal are called ***the quantization levels***, whereas the distance between two successive quantization levels is called ***the quantization step size*** ***or*** ***resolution***.

Sec. 1.6 Cont. - 26

Table1.2 Numerical Illustration of Quantization with one significant digit using Truncation or Rounding

n	x(n) Discrete-time signal	$x_q(n)$ (Truncation)	$x_q(n)$ (Rounding)	$e_q(n) = x_q(n) - x(n)$ (Rounding)
0	1	1.0	1.0	0.0
1	0.9	0.9	0.9	0.0
2	0.81	0.8	0.8	-0.01
3	0.729	0.7	0.7	-0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	-0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	-0.03046721
9	0.387420489	0.3	0.4	0.012579511

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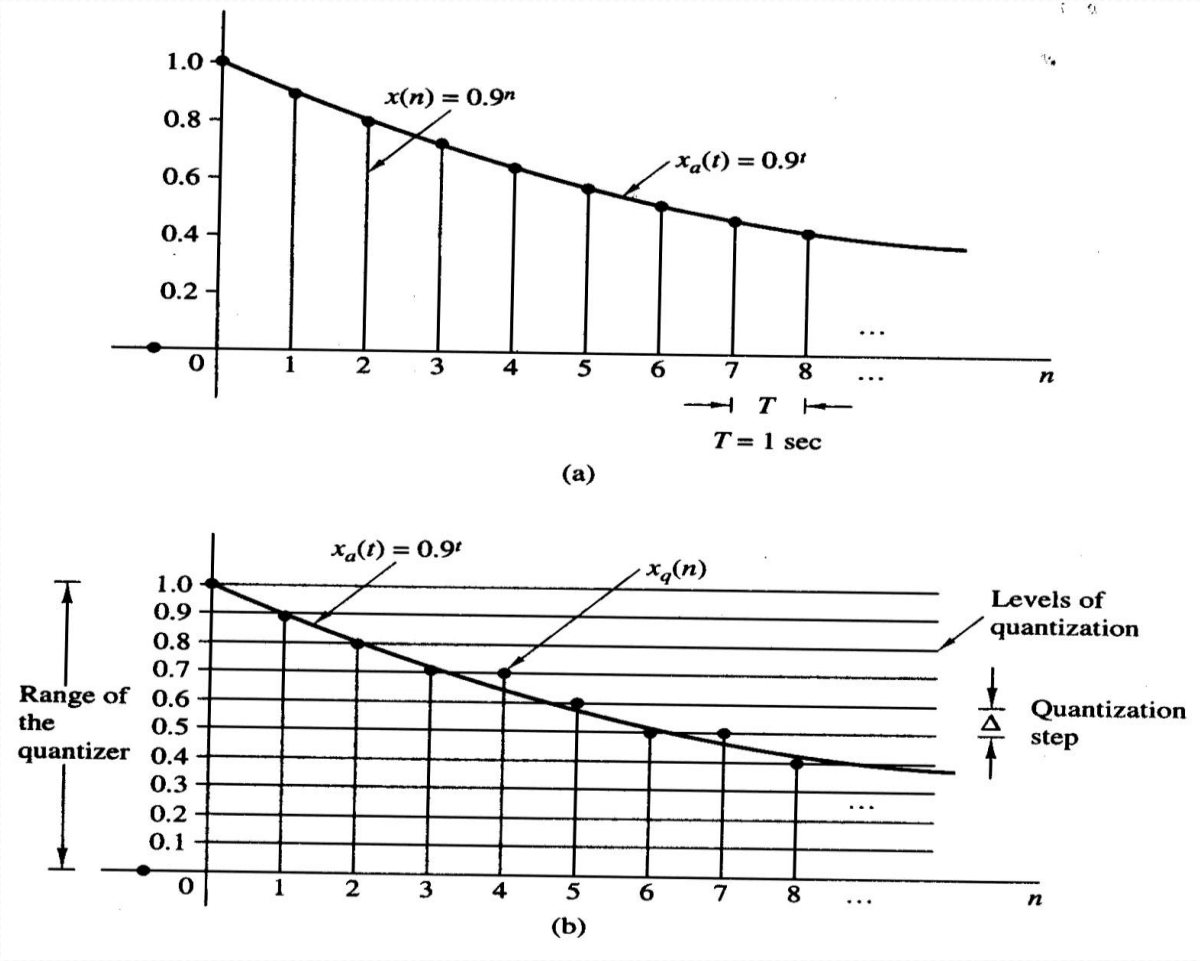


Fig. 1.15 Illustration of quantization

Sec. 1.6 Cont. - 28

- The rounding quantizer assigns each sample of x(n) to the nearest quantization level.

- In contrast, a quantizer that performs truncation would have assigned each sample of x(n) to the quantization level below it.

- The quantization error $e_q(n)$ in rounding is limited to the range of - $\Delta/2$ to $\Delta/2$, i.e.,

$$-\frac{\Delta}{2} \le e_q(n) \le \frac{\Delta}{2} \qquad (1.54)$$

- **In other words,** the instantaneous quantization error **cannot exceed half of the quantization step** (see **Table (1.2)**).

Sec. 1.6 Cont. - 29

- If x_{min} and x_{max} represent the minimum and maximum value of x (n) and L is the number of quantization levels, then

$$\Delta = \frac{x_{\max} - x_{\min}}{L - 1} \qquad (1.55)$$

- We define the dynamic range of the signal as $x_{max} - x_{min} = D\ R$
- In our example we have $x_{max} = 1$, $x_{min} = 0$, and $L = 11$, which leads to $\Delta = 0.1$.

- Note that if the dynamic range is fixed, increasing the number of quantization levels, L results in a decrease of the quantization step size. Thus the quantization error decreases and the accuracy of the quantizer increases.

- In practice we can reduce the quantization error to an insignificant amount by choosing a sufficient number of quantization levels.

Sec. 1.6 Cont. - 30

1.6.4 Coding of Quantized Samples

-The coding process in an A/D converter assigns a unique binary number to each quantization level.

-If we have **L** levels we need at least **L** different binary numbers.

-With a word length of **b** bits we can create 2^b different binary numbers.

- Hence we have $2^b \geq L$, **or** equivalently, $b \geq \log_2 L$.

-Thus the number of bits required in the coder is the smallest integer greater than or equal to $\log_2 L$.($b \geq \log_2 L$)

Sec. 1.6 Cont. - 31

- **In our example** it can easily be seen that we need a coder with b = 4 bits.

- In our example we have x_{max} =1, x_{min} = 0, and L = 11, which leads to step size = 0.1.

- **Commercially** available A/D converters may be obtained with finite precision of b = 16 or less.

- **Generally,** the higher the sampling speed and the finer the quantization, the more expensive the device becomes.

Sec. 1.7

1.7 DIGITAL-TO-ANALOG CONVERSION

- To convert a digital signal into an analog signal we can use a digital-to-analog (D/A) converter.
- As stated previously, the task of a D/A converter is to interpolate between samples.

- The sampling theorem specifies the optimum interpolation for a band limited signal.

- However, this type of interpolation is too complicated and, hence impractical, as indicated previously.

- From a practical viewpoint, the simplest D/A converter is the zero-order hold shown in **Fig. 1.13**, which simply **holds constant the value of one sample until the next one is received.**

Sec. 1.7 Cont. - 1

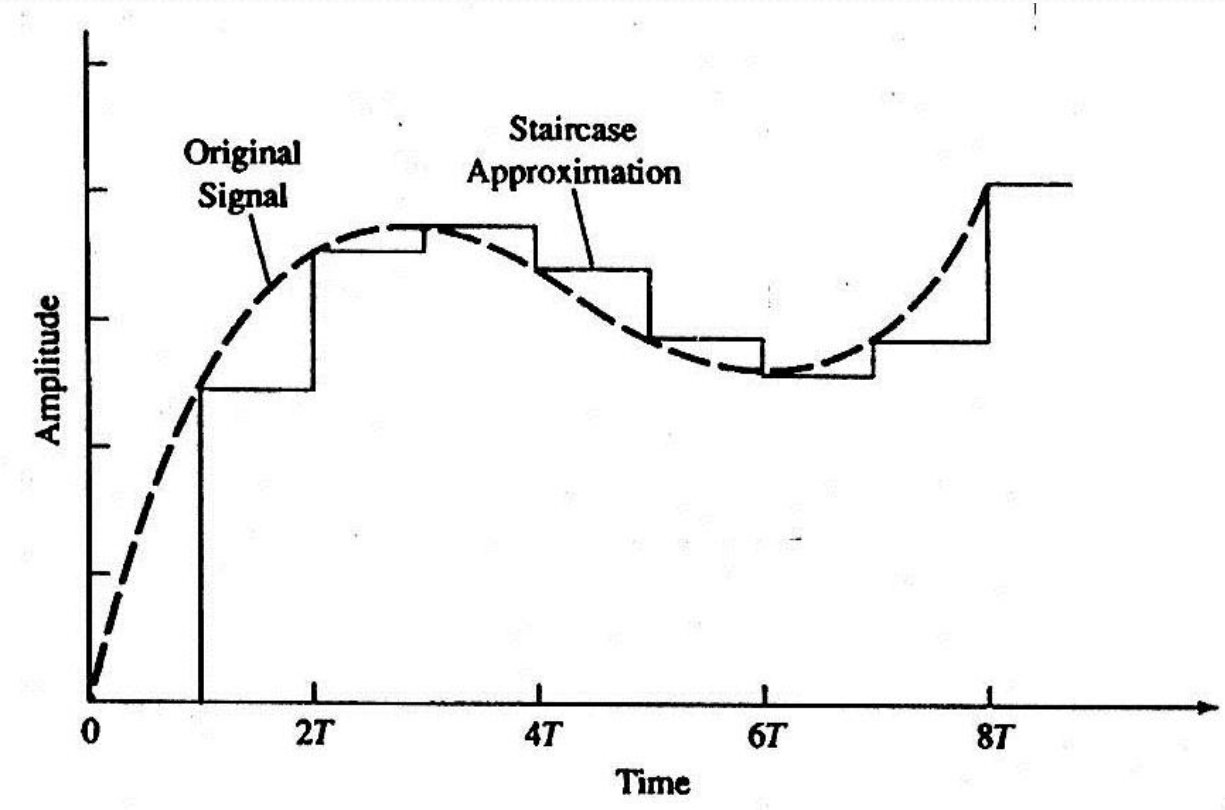


Fig .1.13 Zero-order hold digital-to-analog (D/A) conversion.

Sec. 1.7 Cont. - 2

- Additional improvement can be obtained by **using linear interpolation as shown** in **Fig. 1.14** to connect successive samples with straight-line segments.

- Better interpolation can be achieved by using more sophisticated higher-order interpolation techniques.

Sec. 1.7 Cont. - 3

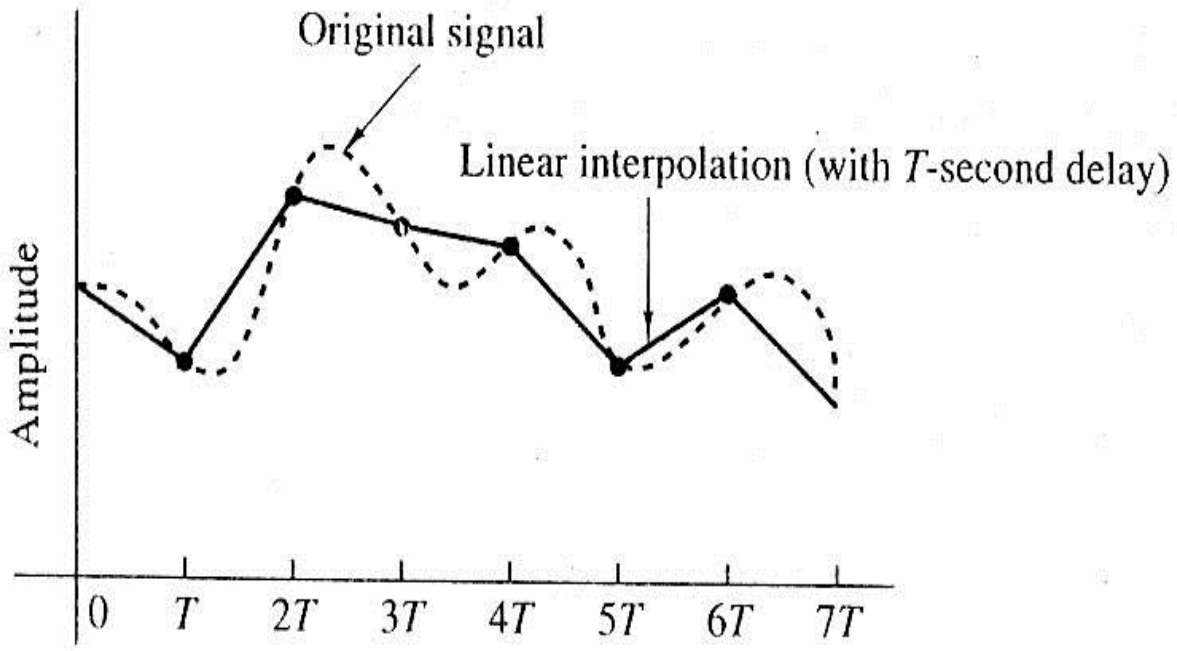


Fig. 1.14 Linear point connector (with T-second delay)

Chapter 1 - Examples

Example 1.1

Consider the analog signal $x_a(t) = 3 \cos 100\pi t$

a- Determine the minimum sampling rate required to avoid aliasing.

b- Suppose that the signal is sampled at the rate $F_s = 200$ Hz. What is the discrete-time signal obtained after sampling?

c- Suppose that the signal is sampled at the rate $F_s = 75$Hz. What is the discrete-time signal obtained after sampling?

d- What is the frequency $0 < f < F_s/2$ of sinusoid that yields samples identical to those obtained in part (c) ?

Chapter 1– Examples- Cont- 1

Solution

a- The frequency of the analog signal is f = 50 Hz.
Hence the minimum sampling rate required to avoid aliasing is F_s = 100 Hz.

b- If the signal is sampled at Fs = 200 Hz, the discrete-time signal is

$$x(n) = 3\cos\frac{100\pi}{200}n = 3\cos\frac{\pi}{2}n$$

c- If the signal is sampled at F_s = 75 Hz, the discrete-time signal is:

$$x(n) = 3\cos\frac{100\pi}{75}n = 3\cos\frac{4\pi}{3}n = 3\cos(2\pi - \frac{2\pi}{3})n = 3\cos\frac{2\pi}{3}n$$

Chapter 1– Examples- Cont- 2

d- For the sampling rate of $F_s = 75$ Hz, we have: $f = F\,F_s = 75F$

The frequency of the sinusoid in part (c) is $F = \frac{1}{3}$.

Hence f = 25 Hz .

Clearly, the sinusoidal signal

$$y_a(t) = 3\cos 2\pi ft = 3\cos 50\,\pi t$$

sampled at Fs = 75 samples/s yields identical samples.

Hence f = 50 Hz is an alias of f = 25 Hz for the sampling rate $F_s = 75$ Hz.

Chapter 1– Examples- Cont- 3

Example 1.2

Consider the analog signal:

$$x_a(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - \cos 100\pi t$$

what is the Nyquist rate for this signal?

Solution

The frequencies present in the signal above are
$f_1 = 25$ Hz. $f_2 = 150$ Hz, $f_3 = 50$Hz

Thus $f_{max} = 150$Hz and according to (1.47), $F_s > 2f_{max} = 300$ Hz

The Nyquist rate is $f_N = 2f_{max}$.

Hence $f_N = 300$ Hz

Chapter 1– Examples- Cont- 4

Discussion:

- It should be observed that the signal component 10 sin 300 π t, sampled at the Nyquist rate f_N =300, results in the samples 10 sin π n, which are identically zero.

- In other words, we are sampling the analog sinusoid at its zero-crossing points, and hence we miss this signal component completely.

- This situation would not occur if the sinusoid is offset in phase by some amount θ .

- In such a case we have 10 sin (300 π t+ θ) sampled at the Nyquist rate f_N = 300 samples per second, which yields the samples:

$$10 \sin (\pi n + \theta) = 10(\sin \pi n \cos \theta + \cos \pi n \sin \theta)$$

$$= 10 \text{ Sin } \theta \cos \pi n$$

$$= (-1)^n 10 \sin \theta$$

Chapter 1– Examples- Cont- 5

- Thus if $\theta \neq 0$ or π , the samples of the sinusoid taken at the Nyquist rate are not all zero.

- However, we still cannot obtain the correct amplitude from the samples when the phase θ is unknown.

- A simple remedy that avoids this potentially troublesome situation is to sample the analog signal *at a rate higher than the Nyquist rate.*

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Example 1.3

Consider the analog signal
$x_a(t) = 3 \cos 2000\pi\, t + 5 \sin 6000\pi\, t + 10 \cos 12000\pi\, t$

(a) What is Nyquist rate for this signal?

(b) Assume now that we sample this signal using a sampling rate $F_s = 5000$ samples/s. What is the discrete-time signal obtained after sampling?

(c) What is the analog signal $y_a(t)$ we can reconstruct from the samples if we use ideal interpolation?

Chapter 1– Examples- Cont- 7

Solution

(a) The frequencies existing in the analog signal are

$$f_1 = 1 \text{ kHz}, \quad f_2 = 3 \text{ kHz}, \quad f_3 = 6 \text{ kHz}$$

Thus $f_{max} = 6$ kHz, and according to the sampling theorem,

$$F_s > 2f_{max} = 12 \text{ kHz}$$

The Nyquist rate is

$$f_N = 12 \text{ kHz}$$

Chapter 1– Examples- Cont- 8

(b) Since we have chosen F_s= 5 kHz, the folding frequency is:

$$\frac{F_s}{2} = 2.5 \text{ kHz}$$

and this is the maximum frequency that can be represented uniquely by the sampled signal.

By making use of (1-30) we obtain:

$$x(n) = x_a(nT) = x_a\left(\frac{n}{F_s}\right)$$

$$= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(\frac{3}{5}\right)n + 10\cos 2\pi\left(\frac{6}{5}\right)n$$

$$= 3\cos\left(2\pi\left(\frac{1}{5}\right)n\right) + 5\sin\left(2\pi\left(1-\frac{2}{5}\right)n\right) + 10\cos\left(2\pi\left(1+\frac{1}{5}\right)n\right)$$

$$= 3\cos 2\pi\left(\frac{1}{5}\right)n + 5\sin 2\pi\left(-\frac{2}{5}\right)n + 10\cos 2\pi\left(\frac{1}{5}\right)n$$

Finally, we obtain: $x(n) = 13\cos 2\pi\left(\frac{1}{5}\right)n - 5\sin 2\pi\left(\frac{2}{5}\right)n$

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(c) Since only the frequency components at 1 kHz and 2 kHz are present in the sampled signal, the analog signal we can recover is

$$y_a(t) = 13 \cos 2000\pi t - 5 \sin 4000\pi t$$

which is obviously different from the original signal $x_a(t)$.
This distortion of the original analog signal was caused by the aliasing effect, due to the low sampling rate used.

Chapter 1– Examples- Cont- 10

Example 1.4: DISSCUSION SHOWS THE ALIASING CONCEPT

The implications of these frequency relations can be fully appreciated by considering the two analog sinusoidal signals

$$x_1(t) = \cos 2\pi (10)t$$

$$x_2(t) = \cos 2\pi (50)t \qquad (1)$$

which are sampled at a rate $F_s = 40$ Hz.

The corresponding discrete-time signals or sequences are:

$$x_1(n) = \cos 2\pi(\tfrac{1}{40})n = \cos \tfrac{\pi}{2}n$$

$$x_2(n) = \cos 2\pi(\tfrac{50}{40})n = \cos \tfrac{5\pi}{2}n \qquad (2)$$

Chapter 1– Examples- Cont- 11

However, $\cos 5\pi n/2 = \cos(2\pi n + \pi n/2) = \cos \pi n/2$.

Hence $x_2(n) = x_1(n)$.

Thus the sinusoidal signals are identical and, consequently, indistinguishable.

If we are given the sampled values generated by $\cos(\pi/2)\, n$, there is some ambiguity as to whether these sampled values correspond to $x_1(t)$ or $x_2(t)$

Since $x_2(t)$ yields exactly the same values as $x_1(t)$ when the two are sampled at $F_s = 40$ samples per second, we say that the frequency $f_2 = 50$ Hz is an alias of the frequency $f_1 = 10$ Hz at the sampling rate of 40 samples per second.

It is important to note that f_2 is not the only alias of f_1.

In fact at the sampling rate of 40 samples per second, the frequency $f_3 = 90$ Hz is also an alias of f_1, as is the frequency $f_4 = 130$ Hz, and so on.